Time series analysis

AB–204 Arctic population ecology, October 27–30, 2026

J.-Prof. Dr. Timo Adam

Course overview

Visualising (time series) data
October 27, 2026

- what is a time series?
- R and RStudio
- scatter and time series plots
- boxplots, histograms, densities

Basics of linear regression
October 28, 2026

- statistical models
- simple and multiple linear regression
- interpreting effects
- residuals and model selection

Trend, seasonal, and cyclical effects
October 29, 2026

- time series components
- trend and seasonality
- autocorrelation (ACF, PACF)
- autoregressive models

Summary and outlook
October 30, 2026

- summary of the course
- outlook: other extensions of linear regression
- your questions

Questions? menti.com/al94rhvnrre9.

Lecture 3: Trend, seasonal, and cyclical effects

3.1 Introduction — time series components

3.2 Linear regression with trend

3.3 Autocorrelation

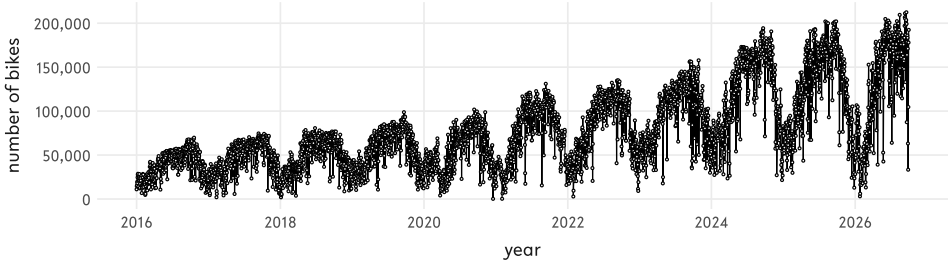
3.4 Linear regression with seasonal effects

3.5 Cyclical effects and autoregressive models

Questions? menti.com/al94rhvnrre9.

3.1 Introduction — example 1: bike sharing

Daily number of rented bikes in New York City, January 2016 – September 2026[1]:

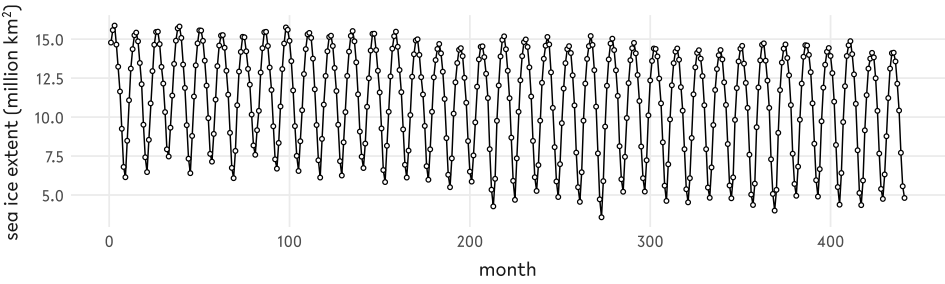


What patterns can you identify?

[1] Data from https://citibikenyc.com/system-data.

3.1 Introduction — example 2: Arctic sea ice

Monthly Arctic sea ice extent, January 1990 – September 2026[2]:

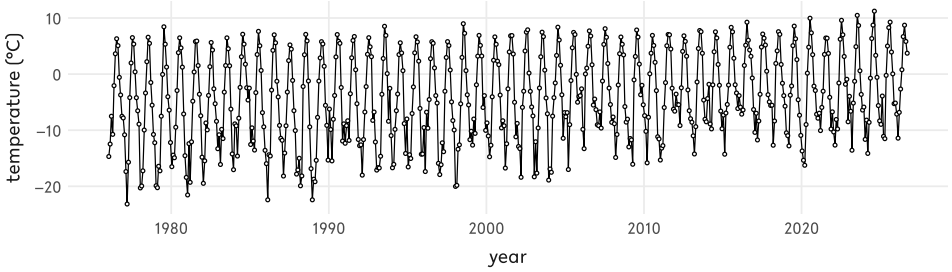


What patterns can you identify?

[2]Data from the Sea Ice Index of the National Snow and Ice Data Center, https://nsidc.org/data/seaice_index; monthly values from NOAA, https://www.ncei.noaa.gov/access/monitoring/snow-and-ice-extent.

3.1 Introduction — example 3: temperature in Longyearbyen

Monthly mean temperature at Svalbard Airport, January 1976 – September 2026[3]:



What patterns can you identify?

[3] Data from NOAA's Global Historical Climatology Network (station Svalbard Airport, SV000001008), doi:10.7289/V5D21VHZ; monthly means of the daily mean of the minimum and maximum temperature; in the course data set `svalbard_temperature.csv`.

3.1 Introduction — the additive component model

Time series are often made up of different **components**:

The **additive time series component model** is given by

$$Y_t = T_t + S_t + C_t + \varepsilon_t,$$

where

- T_t is the **trend** (the long-term direction);
- S_t is the **seasonal component** (a pattern that repeats with a fixed period, e.g., every year);
- C_t is the **cyclical component** (ups and downs without a fixed period);
- ε_t is the **remainder** (random noise, which cannot be predicted).

Today: model T_t, S_t, and C_t using linear regression — until only random noise is left.

3.1 Introduction — examples

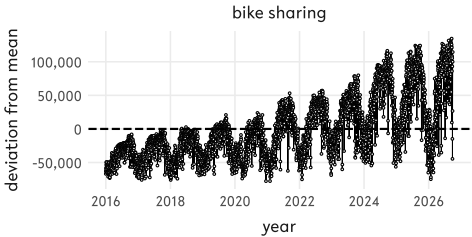
- **Trend**: long-term population growth or decline; long-term change in temperature, in the demand for bikes, or in sea ice extent.
- **Seasonal component**: fluctuations of population size, temperature, or sea ice extent **within a year**; fluctuations of the demand for bikes within a week or year.
- **Cyclical component**: fluctuations without a fixed period, e.g., population cycles of predators and prey, or El Niño.
- **Remainder**: ideally completely random — otherwise, there's still something left that can be modelled!

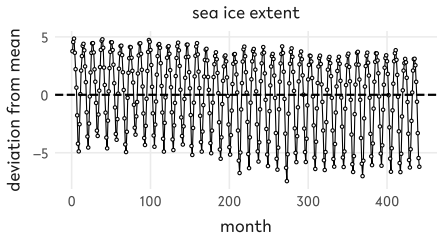
Seasonal effects are **predictable** (we know when winter comes); cyclical effects are not (we don't know exactly when the next lynx peak comes).

3.1 Introduction — visualising the trend

Each value minus the overall mean, plotted over time:

```
bikes <- read.csv("bike_sharing.csv"); bikes$date <- as.Date(bikes$date)
plot(x = bikes$date, y = bikes$count - mean(bikes$count, na.rm = TRUE),
     type = "b", cex = 0.3,
     xlab = "year", ylab = "deviation from mean")
abline(h = 0, lty = 2) # long-term mean
```





Bikes: more and more bikes over time (trend), and more in summer (seasonal component).

Sea ice: the seasonal component dominates; is there a trend?

Lecture 3: Trend, seasonal, and cyclical effects

3.1 Introduction — time series components

3.2 Linear regression with trend

3.3 Autocorrelation

3.4 Linear regression with seasonal effects

3.5 Cyclical effects and autoregressive models

3.2 Linear regression with trend — linear trend

A **linear trend** is a straight line over time — a linear model with the time index $t = 1, 2, 3, \ldots$ as covariate:

$$\mathbb{E}(\text{number of bikes}_t) = \beta_0 + \beta_1 \cdot t.$$

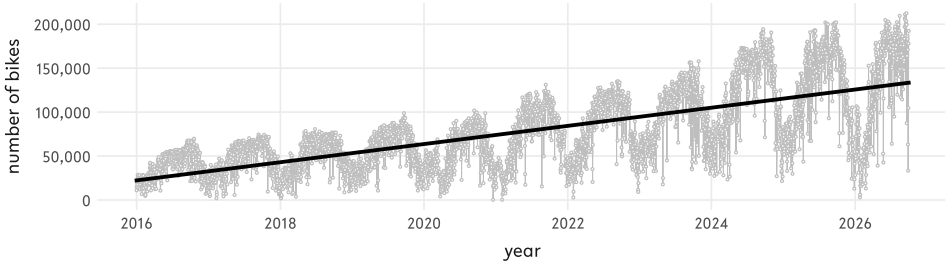
```
bikes$t <- 1:nrow(bikes) # time index: 1, 2, ..., 3926
mod_lin <- lm(formula = count ~ t, data = bikes)
summary(mod_lin)
```

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 2.227e+04  1.051e+03   21.18   <2e-16 ***
t           2.830e+01  4.633e-01   61.09   <2e-16 ***
```

Every day, the expected number of bikes increases by $\hat{\beta}_1 = 28.3$ (10,330 per year). As $p < 0.05$, the trend is statistically significant.

3.2 Linear regression with trend — linear trend

```
plot(x = bikes$date, y = bikes$count, type = "b", cex = 0.3, col = "grey",
     xlab = "year", ylab = "number of bikes")
lines(x = bikes$date, y = predict(mod_lin, newdata = bikes), lwd = 2)
```



The increase seems to speed up in recent years — is a straight line flexible enough?

3.2 Linear regression with trend — polynomial trends

More flexible trends: curves instead of a straight line — **quadratic** or **cubic** terms of t:

$$\mathbb{E}(\text{number of bikes}_t) = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot t^2,$$

$$\mathbb{E}(\text{number of bikes}_t) = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot t^2 + \beta_3 \cdot t^3.$$

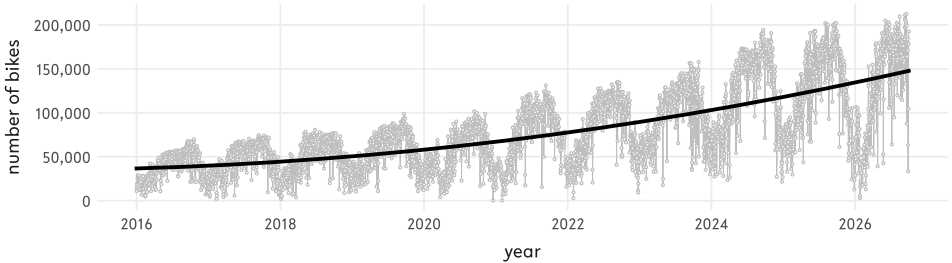
```
mod_quad <- lm(formula = count ~ t + I(t^2), data = bikes) # I() for calculations
mod_cub <- lm(formula = count ~ t + I(t^2) + I(t^3), data = bikes)
```

Model selection using the AIC (Lecture 2):

```
AIC(mod_lin) # 92557.6
AIC(mod_quad) # 92409.1
AIC(mod_cub) # 92409.5
```

The AIC prefers the quadratic trend (the cubic term doesn't improve the fit enough).

3.2 Linear regression with trend — quadratic trend



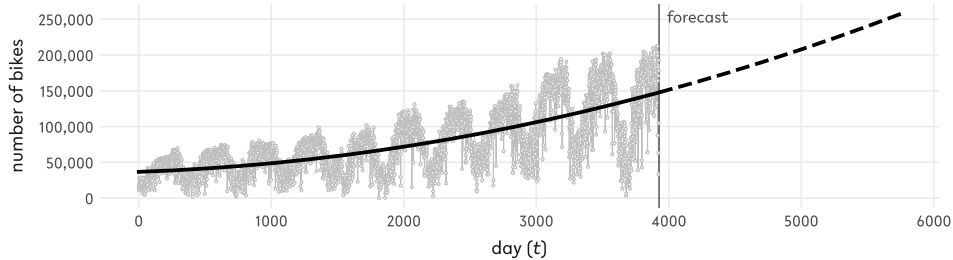
The quadratic trend captures the faster increase in recent years.

But the seasonal pattern is still there — we'll come back to this in 3.4.

3.2 Linear regression with trend — forecasting

Forecasts: predict for **future** values of t — here the next five years:

```
future <- data.frame(t = 3927:(3926 + 5 * 365)) # the next five years
predict(mod_quad, newdata = future) # forecasts
```



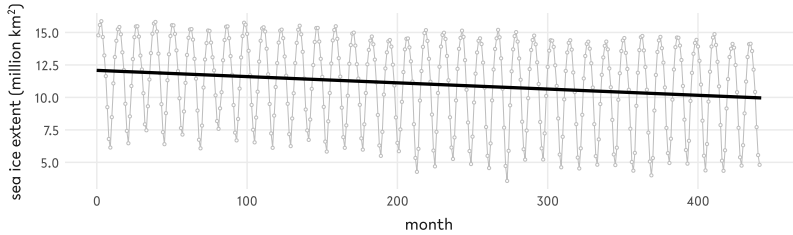
The quadratic trend predicts about 258,000 bikes per day in five years — plausible? Be careful when predicting beyond the data (**extrapolation**): curves can go in strange directions.

3.2 Linear regression with trend — the sea ice data

The same for the sea ice extent:

```
ice <- read.csv("sea_ice.csv")
ice$t <- 1:nrow(ice) # time index: 1, 2, ..., 441 (months)
mod_ice <- lm(formula = extent ~ t, data = ice)
```

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 12.081521   0.313312  38.561  < 2e-16 ***
t           -0.004797   0.001228  -3.905 0.000109 ***
```



Every month, the expected sea ice extent decreases by 0.0048 million km^2 (0.058 per year).

Lecture 3: Trend, seasonal, and cyclical effects

3.1 Introduction — time series components

3.2 Linear regression with trend

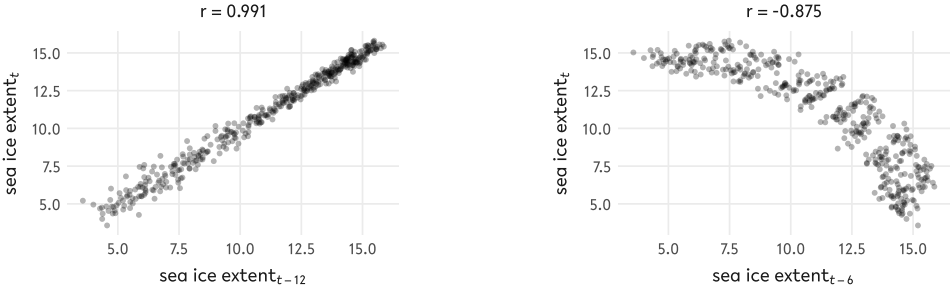
3.3 Autocorrelation

3.4 Linear regression with seasonal effects

3.5 Cyclical effects and autoregressive models

3.3 Autocorrelation — correlation with the past

In time series, current observations are correlated with **past** observations:

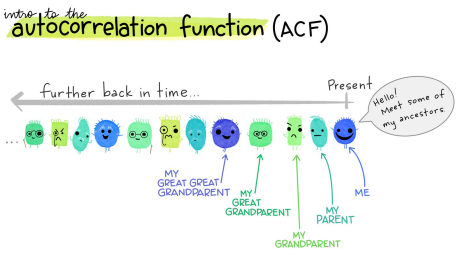


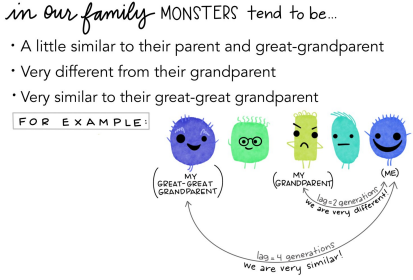
- Sea ice extent now vs. 12 months ago: $r = 0.991$ — almost the same.
- Sea ice extent now vs. 6 months ago: $r = -0.875$ — summer follows winter.

The correlation of a time series with its own past (lagged) values is called **autocorrelation**.

3.3 Autocorrelation — a non-technical introduction

What is autocorrelation?[4]



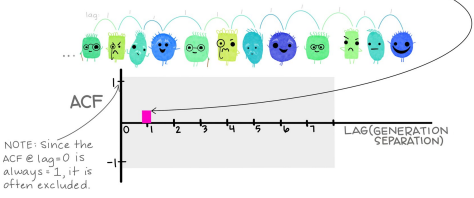


[4] Artwork by Allison Horst, https://allisonhorst.com/time-series-acf.

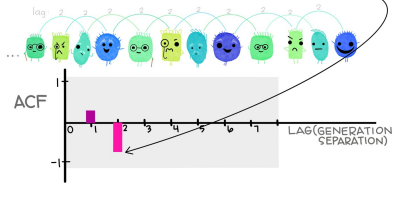
3.3 Autocorrelation — a non-technical introduction

What is autocorrelation?

At lag =1, we find the correlation between **monsters** and their **parent.**
They are somewhat positively correlated.



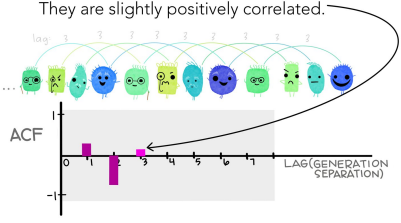
At lag = 2, we find the correlation between **monsters** and their **grandparent.**
Since they tend to be very different, we find a negative correlation at lag = 2.



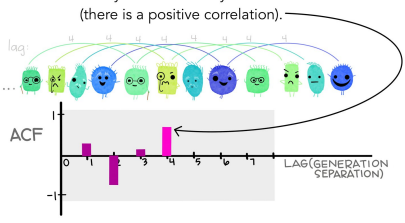
3.3 Autocorrelation — a non-technical introduction

What is autocorrelation?

At lag = 3, we find the correlation between **monsters** and their **great-grandparent**. They are slightly positively correlated.



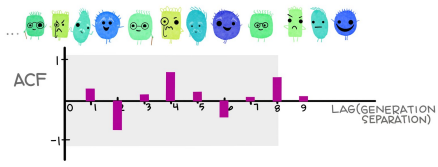
At lag = 4, we find the correlation between **monsters** and their **great-great grandparent**. They tend to be very similar (there is a positive correlation).



3.3 Autocorrelation — a non-technical introduction

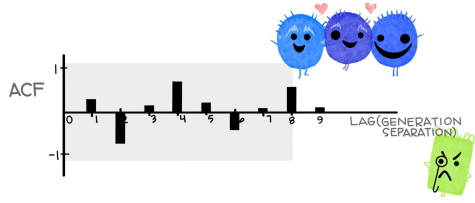
What is autocorrelation?

...and we continue finding the correlations as we increase the lag (generations) between the monsters...



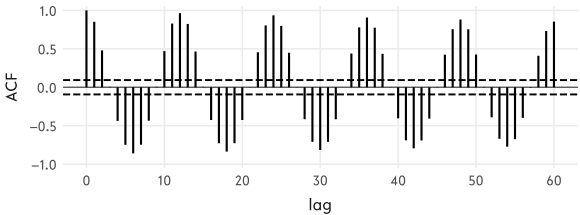
in Summary:

The autocorrelation function (ACF) tells us the correlation between observations and those that came before them, separated by different lags (here, monster generations)!



3.3 Autocorrelation — the autocorrelation function (ACF)

- **Autocorrelation at lag** k: correlation between observations that are k time steps apart (e.g., lag 12: the same month one year earlier).
- The **autocorrelation function** (ACF) shows the autocorrelation for many lags $k = 0, 1, 2, \ldots$ (at lag 0 always 1).
- **Dashed lines**: values outside them show statistically significant autocorrelation.



ACF of the sea ice extent:

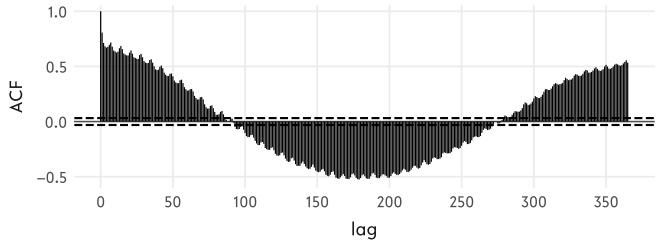
- positive at lags 12, 24, 36, ...;
- negative at lags 6, 18, 30, ...

A clear **seasonal** pattern with a period of 12 months.

3.3 Autocorrelation — the ACF in R

ACF of the bike data **after removing the trend** — i.e., of the residuals of the quadratic trend model:

```
acf(residuals(mod_quad), lag.max = 365, ci.col = "black",
    main = "", xlab = "lag", ylab = "ACF")
```

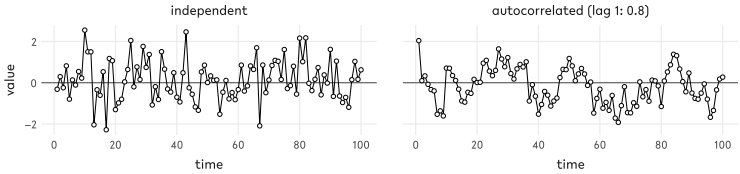


Significant autocorrelation at almost all lags: days close in time (or one year apart) are similar, days half a year apart are opposite.

The residuals are **not** independent: the trend alone doesn't capture the seasonal pattern.

3.3 Autocorrelation — why does it matter?

The linear model assumes **independent** errors (Lecture 2) — for time series, this is often violated:



- Autocorrelated observations carry **overlapping information**: today tells us a lot about tomorrow.
- 3,916 autocorrelated days contain less information than 3,916 independent days — here, only about as much as 420 independent days.
- Ignoring this makes `lm()` **overconfident**: it thinks it knows more than it does — standard errors, confidence intervals, and p-values are too small.
- The remedy: model the dependence — seasonal components and covariates (3.4), autoregressive models (3.5).

Lecture 3: Trend, seasonal, and cyclical effects

3.1 Introduction — time series components

3.2 Linear regression with trend

3.3 Autocorrelation

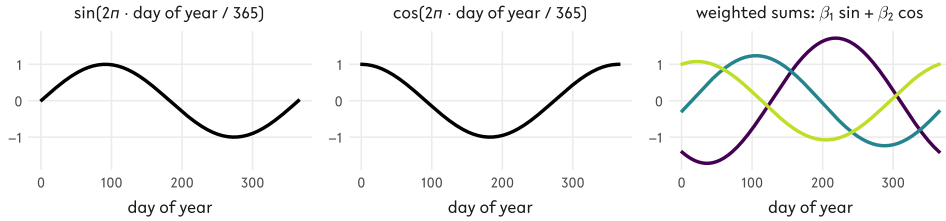
3.4 Linear regression with seasonal effects

3.5 Cyclical effects and autoregressive models

3.4 Linear regression with seasonal effects — sine and cosine

A smooth seasonal pattern using **sine and cosine waves**:

$$S_t = \beta_1 \cdot \sin\left(\frac{2 \cdot \pi \cdot \text{day of year}_t}{365}\right) + \beta_2 \cdot \cos\left(\frac{2 \cdot \pi \cdot \text{day of year}_t}{365}\right).$$



- Sine and cosine both repeat every 365 days.
- Combined, they give a wave with any timing of the **peak** and any **height** — both estimated from the data.

3.4 Linear regression with seasonal effects — the bike sharing data

Trend plus seasonal component:

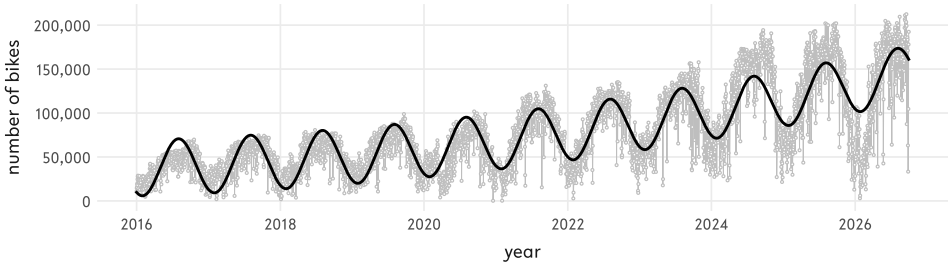
$$\mathbb{E}(\text{number of bikes}_t) = \underbrace{\beta_0 + \beta_1 \cdot t + \beta_2 \cdot t^2}_{\text{trend}} + \underbrace{\beta_3 \cdot \sin\left(\frac{2\pi \cdot \text{day of year}_t}{365}\right) + \beta_4 \cdot \cos\left(\frac{2\pi \cdot \text{day of year}_t}{365}\right)}_{\text{seasonal component}}.$$

```
mod_season <- lm(formula = count ~ t + I(t^2) + sin(2 * pi * day_of_year / 365)
                 + cos(2 * pi * day_of_year / 365), data = bikes)
summary(mod_season)
```

```
Coefficients:
                                Estimate Std. Error t value Pr(>|t|)
(Intercept)                     3.729e+04  1.110e+03  33.591   <2e-16 ***
t                               6.986e+00  1.303e+00   5.360 8.78e-08 ***
I(t^2)                          5.182e-03  3.210e-04  16.143   <2e-16 ***
sin(2 * pi * day_of_year/365) -1.623e+04  5.206e+02 -31.175   <2e-16 ***
cos(2 * pi * day_of_year/365) -2.725e+04  5.204e+02 -52.356   <2e-16 ***
```

3.4 Linear regression with seasonal effects — the bike sharing data

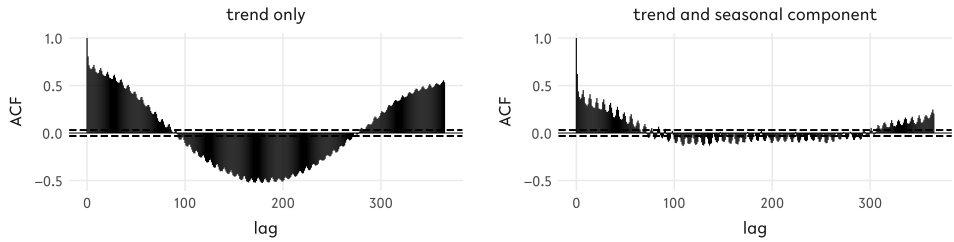
```
plot(x = bikes$date, y = bikes$count, type = "b", cex = 0.3, col = "grey",
     xlab = "year", ylab = "number of bikes")
lines(x = bikes$date, y = predict(mod_season, newdata = bikes), lwd = 2)
```



Trend and seasonal component together. The AIC drops from 92,409.1 (quadratic trend only) to 89,775.2, and R^2 increases from 0.51 to 0.75.

3.4 Linear regression with seasonal effects — what's left?

ACF of the residuals, before and after adding the seasonal component (365 lags):



Much better: the seasonal pattern is mostly gone.

But there's still autocorrelation at short lags (e.g., weather lasting several days) — more covariates, or a cyclical component, could help.

3.4 Linear regression with seasonal effects — months as categories

Alternatively: one level per month — the month as a **categorical** covariate (as species in Lecture 2):

$$\mathbb{E}(\text{sea ice extent}_t) = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot \mathbb{1}_{\text{month}_t=1} + \beta_3 \cdot \mathbb{1}_{\text{month}_t=2} + \ldots + \beta_{12} \cdot \mathbb{1}_{\text{month}_t=11},$$

where $\mathbb{1}_{\text{month}_t=i}$ is 1 in month *i* and 0 otherwise (December is the reference category).

```
ice$month <- relevel(as.factor(ice$month), ref = "12") # December as reference
mod_ice_month <- lm(formula = extent ~ t + month, data = ice)
summary(mod_ice_month)
```

- Each month gets its own level — any seasonal shape is possible.
- Needs 11 coefficients (instead of 2 for sine and cosine) — fine for monthly data, but too many for days of the year.

3.4 Linear regression with seasonal effects — months as categories

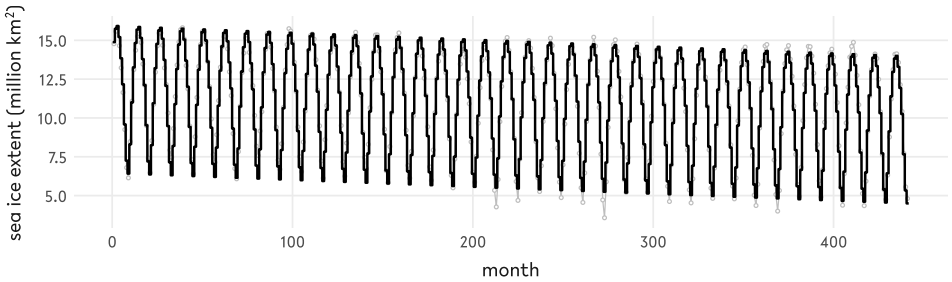
```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 13.314297   0.077010 172.891  < 2e-16 ***
t           -0.004435   0.000155 -28.666  < 2e-16 ***
month1       1.569995   0.096818  16.216  < 2e-16 ***
month2       2.440916   0.096817  25.212  < 2e-16 ***
month3       2.616162   0.096816  27.022  < 2e-16 ***
month4       1.912219   0.096815  19.751  < 2e-16 ***
month5       0.553951   0.096815   5.722 1.98e-08 ***
month6      -1.117560   0.096815 -11.543  < 2e-16 ***
month7      -3.699612   0.096815 -38.213  < 2e-16 ***
month8      -6.027068   0.096815 -62.253  < 2e-16 ***
month9      -6.862093   0.096816 -70.878  < 2e-16 ***
month10     -4.966092   0.097476 -50.947  < 2e-16 ***
month11     -2.263602   0.097476 -23.222  < 2e-16 ***
```

Interpretation:

- In March, the expected sea ice extent is 2.62 million km^2 larger than in December (the maximum).
- In September, it is 6.86 million km^2 smaller than in December (the minimum).
- For a given month, the extent decreases by 0.0044 million km^2 per month (0.053 per year).

3.4 Linear regression with seasonal effects — months as categories

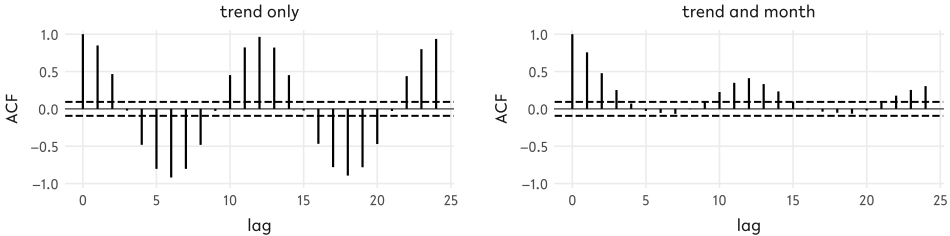
```
plot(x = ice$t, y = ice$extent, type = "b", cex = 0.5, col = "grey",
     xlab = "month", ylab = "sea ice extent")
lines(x = ice$t, y = fitted(mod_ice_month), lwd = 2)
```



The model captures the seasonal pattern — the AIC drops from 2,304.3 (trend only) to 487.5.

3.4 Linear regression with seasonal effects — what's left?

ACF of the residuals, before and after adding the month (`acf(residuals(mod_ice_month), lag.max = 24)`):

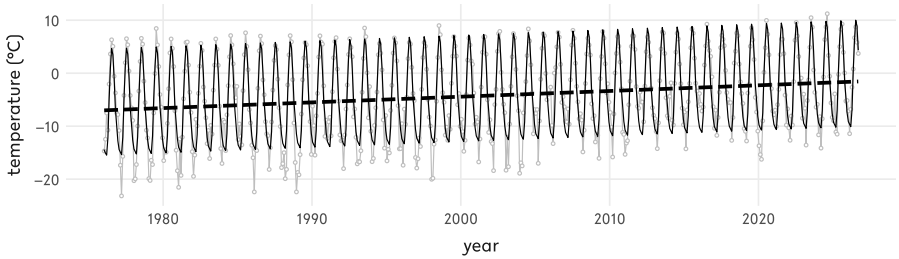


Much better: the seasonal pattern is gone. Some autocorrelation at short lags remains — a cyclical component?

3.4 Linear regression with seasonal effects — is Svalbard warming?

The same model for the temperature in Longyearbyen (Example 3):

```
sv <- read.csv("svalbard_temperature.csv"); sv$t <- 1:nrow(sv)
mod_sv <- lm(formula = temperature ~ t + as.factor(month), data = sv) # January as reference
```

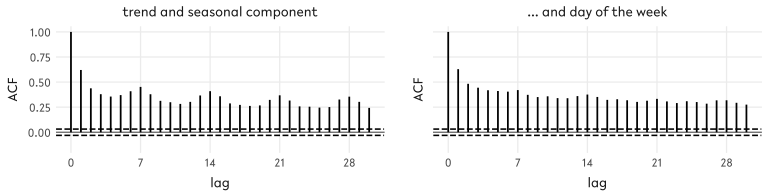


- Within each month, the expected temperature increases by 0.009 °C per month — about **1.1 °C per decade**, or more than 5 °C since 1976 (dashed line).
- The residuals are autocorrelated (0.34 at lag 1) — so the p-value is too small (3.3), but the warming is clear.

3.4 Linear regression with seasonal effects — the weekly pattern

Back to the bikes: the residuals showed small peaks at lags 7 and 14 — a **weekly** pattern. Add the day of the week as a categorical covariate (as species in Lecture 2):

```
mod_week <- lm(formula = count ~ t + I(t^2) + sin(2 * pi * day_of_year / 365)
               + cos(2 * pi * day_of_year / 365) + as.factor(day_of_week), data = bikes)
acf(residuals(mod_week), lag.max = 30, ci.col = "black", main = "")
```



The weekly peaks are gone, and the AIC drops from 89,775.2 to 89,556.5 (most bikes on Wednesdays, fewest on Sundays).

But the autocorrelation at short lags remains (e.g., weather lasting several days) — what about the weather?

3.4 Linear regression with seasonal effects — putting it all together

Trend, season, day of the week, **and** weather — one model for the bike sharing data:

```
mod_all <- lm(formula = count ~ t + I(t^2) + sin(2 * pi * day_of_year / 365)
              + cos(2 * pi * day_of_year / 365) + as.factor(day_of_week)
              + max_temperature + precipitation, data = bikes)
```

model	AIC	R^2	ACF at lag 1
quadratic trend	92,409.1	0.51	0.81
+ season (sine and cosine)	89,775.2	0.75	0.62
+ day of the week	89,556.5	0.76	0.63
+ temperature and precipitation	88,168.9	0.83	0.70

- Each step improves the model: $+1{,}035$ bikes per °F, and $-22{,}441$ bikes per inch of rain (on otherwise comparable days).
- But the residuals remain strongly autocorrelated — today's number of bikes depends on yesterday's. Next: autoregressive models.

Lecture 3: Trend, seasonal, and cyclical effects

3.1 Introduction — time series components

3.2 Linear regression with trend

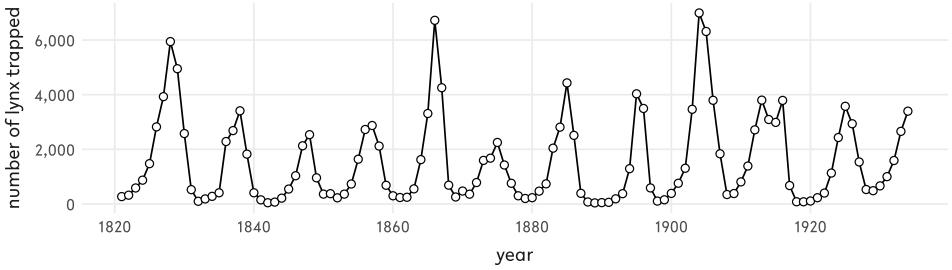
3.3 Autocorrelation

3.4 Linear regression with seasonal effects

3.5 Cyclical effects and autoregressive models

3.5 Cyclical effects — the Canadian lynx

Number of lynx trapped in north-west Canada, 1821–1934[5]:

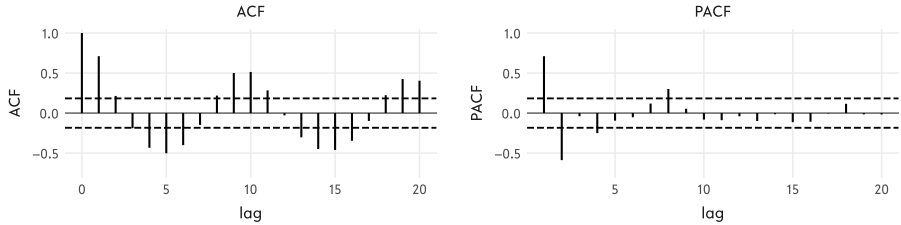


- Regular peaks every 9–10 years, but not with a fixed, known period: a **cyclical** component.
- No obvious trend or seasonal component (annual data).

[5]Data from Campbell, M. and Walker, A. (1977), *Journal of the Royal Statistical Society, Series A*, 140, 411–431, doi:10.2307/2345277; available as `lynx` in R.

3.5 Cyclical effects — partial autocorrelation

- **Chain reaction**: if this year depends on last year, and last year on the year before, then this year is also correlated with two years ago — even without a **direct** effect.
- The **partial autocorrelation** at lag k removes this chain reaction: it measures only the **direct** link between y_t and y_{t-k}.
- The **partial autocorrelation function** (PACF) shows which past years directly affect the current one.



ACF: cycles of about 10 years. PACF: significant at lags 1, 2, 4, and 8 (`acf(lynx)`, `pacf(lynx)`).

3.5 Cyclical effects — autoregressive models

Idea: use **past observations** as covariates.

The **autoregressive model** of order p, AR(p), is given by

$$Y_t = \beta_0 + \beta_1 \cdot y_{t-1} + \beta_2 \cdot y_{t-2} + \ldots + \beta_p \cdot y_{t-p} + \varepsilon_t \quad \Leftrightarrow \quad \mathbb{E}(Y_t) = \beta_0 + \beta_1 \cdot y_{t-1} + \ldots + \beta_p \cdot y_{t-p}.$$

- AR(1): this year's number of lynx depends on last year's number.
- AR(2): it depends on the numbers of the last two years, and so on.
- The PACF suggests the order p: the last lag with significant partial autocorrelation.

(For data with trend or season: model these first, then use AR models for what's left (the residuals).)

3.5 Cyclical effects — autoregressive models in R

```r
mod_ar1 <- arima(lynx, order = c(1, 0, 0)) # AR(1) model
mod_ar1
```

```
Coefficients:
         ar1  intercept
      0.7173  1550.5708
s.e.  0.0649   356.6935

sigma^2 estimated as 1210542:  log likelihood = -960.5,  aic = 1926.99
```

```r
mod_ar2 <- arima(lynx, order = c(2, 0, 0)) # AR(2) model
mod_ar2
```

```
Coefficients:
         ar1      ar2  intercept
      1.1474  -0.5997  1545.4458
s.e.  0.0742   0.0740   181.6736

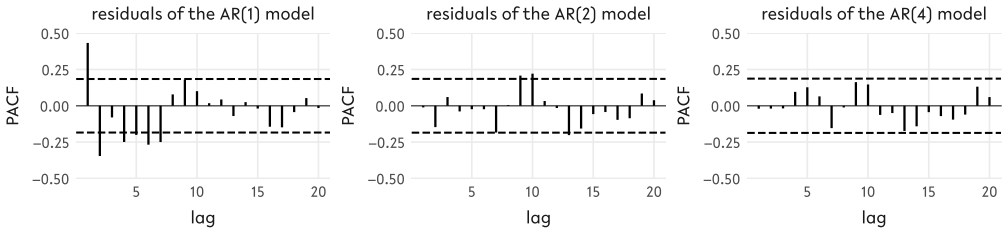
sigma^2 estimated as 768159:  log likelihood = -935.02,  aic = 1878.03
```

(`intercept` is the **mean** of the time series, not β_0.)

3.5 Cyclical effects — checking the residuals

Is there any partial autocorrelation left in the residuals?

```
pacf(residuals(mod_ar1), ci.col = "black", main = "", xlab = "lag", ylab = "PACF")
```



AR(1) and AR(2): there's still significant partial autocorrelation left.

AR(4): no significant partial autocorrelation left (lag 13 is close to the band). The AIC agrees:

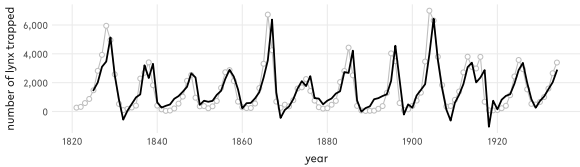
model	AR(1)	AR(2)	AR(3)	AR(4)
AIC	1927.0	1878.0	1880.0	**1874.2**

3.5 Cyclical effects — the AR(4) model

```
mod_ar4 <- arima(lynx, order = c(4, 0, 0)) # AR(4) model
library(lmtest) # install.packages("lmtest") once
coeftest(mod_ar4) # p-values for the coefficients
```

```
z test of coefficients:

             Estimate  Std. Error z value  Pr(>|z|)
ar1          1.124606    0.090328 12.4503 < 2.2e-16 ***
ar2         -0.717374    0.136665 -5.2491 1.528e-07 ***
ar3          0.263374    0.136143  1.9345  0.053046 .
ar4         -0.254328    0.089668 -2.8363  0.004563 **
intercept 1547.385915  136.850051 11.3072 < 2.2e-16 ***
```



Purple: predictions from the four previous years — the AR(4) model reproduces the cycles.

3.5 Cyclical effects — interpretation

What could cause the lynx cycles? **Predator-prey interactions**:

- The main prey of the lynx is the **snowshoe hare**.
- Many lynx in year $t-2$ lead to fewer hares in year $t-1$, and hence to fewer lynx in year t: negative partial autocorrelation at lag 2.
- Many hares cause heavy browsing, which lowers the quality of their food plants in the following years — with effects on the lynx with a lag of about 4 years.

Possibly plant-herbivore effects in addition to predator-prey effects.

Caution with interpretation: these are trapping data, not collected for research (trapping effort may have varied).

Lecture 3 — summary

step	question	in R
trend	how does the level change over time?	`lm(y ~ t + I(t^2))`
forecast	what do we expect in the future?	`predict(mod, newdata = data.frame(t = ...))`
autocorrelation	are the residuals independent?	`acf(residuals(mod))`
seasonal component	is there a pattern within a year?	`sin()` and `cos()`, or `as.factor(month)`
cyclical component	are there cycles of unknown length?	`pacf()`, `arima(y, order = c(p, 0, 0))`
select	which model is best?	`AIC(mod_1, mod_2)`

Table: The main steps of a time series analysis.

Time series = trend + seasonal + cyclical component + remainder. The goal: residuals that look like random noise.

Lecture 3 — summary questions

1. What are the components of the additive component model? Give an ecological example of each.
2. How do you model a trend, and how do you choose between linear, quadratic, and cubic trends? Why are forecasts from polynomial trends risky?
3. What is autocorrelation, and what does the ACF show? What could a spike at lag 7 in the ACF of the residuals of daily data mean?
4. How can a seasonal component be modelled? Why do we need both sine **and** cosine?
5. What is the difference between a seasonal and a cyclical component?
6. What is an AR(p) model, and how does the PACF help you choose p?